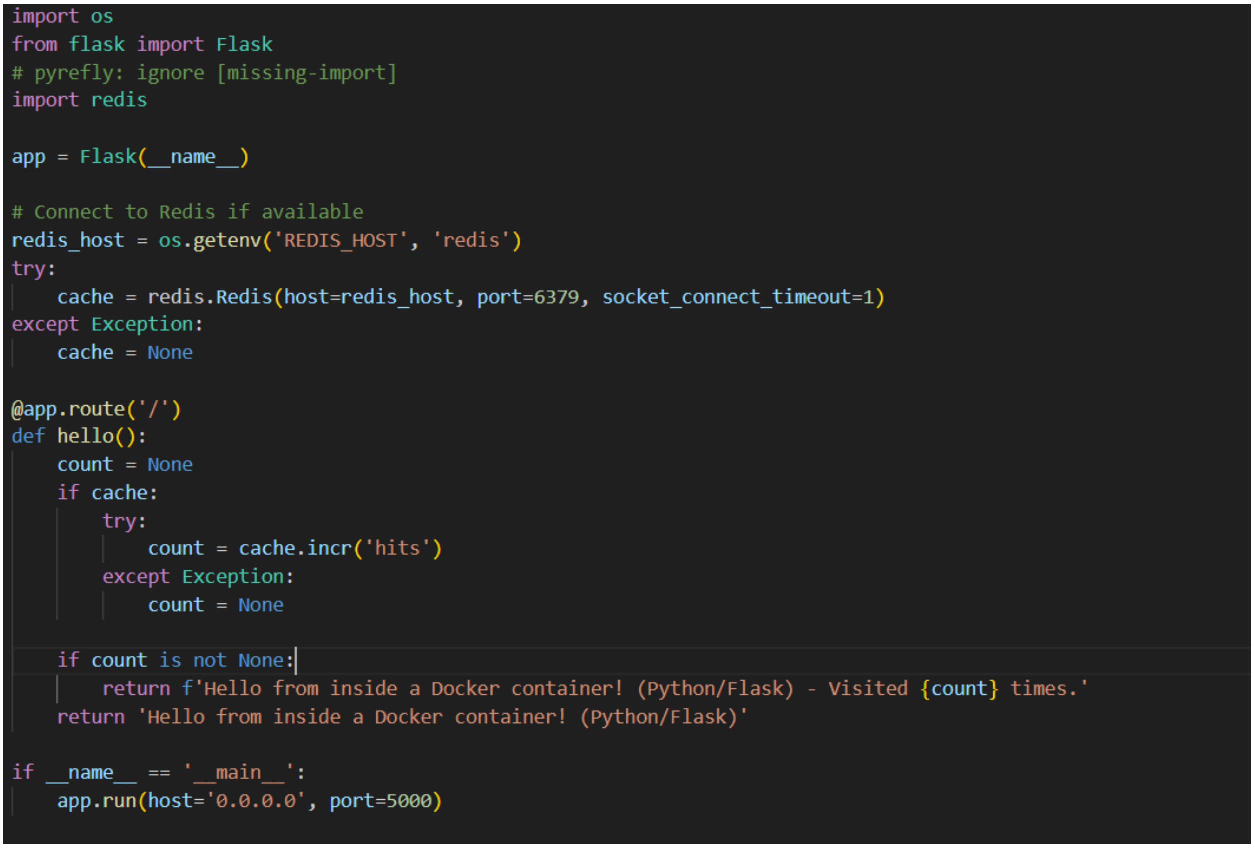
Lab 2: Containerization & Kubernetes Orchestration

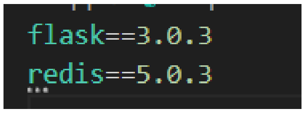
Task 1: Containerize a Sample Application Using Docker

1. app.py



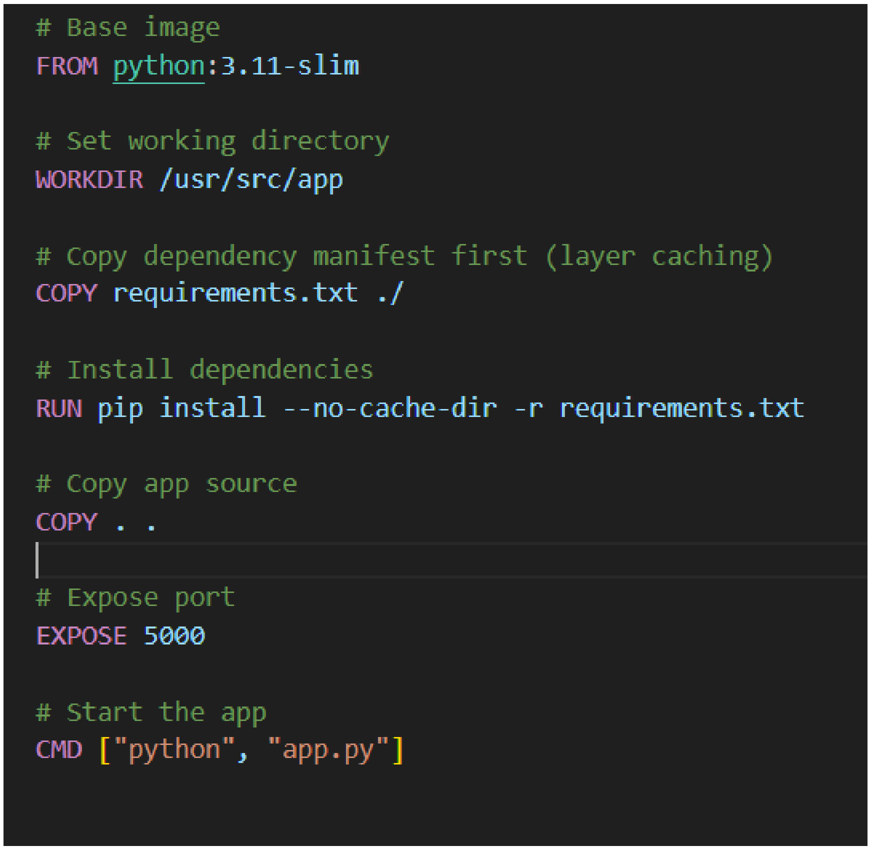
```python
import os
from flask import Flask
# pyrefly: ignore [missing-import]
import redis

app = Flask(__name__)

# Connect to Redis if available
redis_host = os.getenv('REDIS_HOST', 'redis')
try:
    cache = redis.Redis(host=redis_host, port=6379, socket_connect_timeout=1)
except Exception:
    cache = None

@app.route('/')
def hello():
    count = None
    if cache:
        try:
            count = cache.incr('hits')
        except Exception:
            count = None

    if count is not None:
        return f'Hello from inside a Docker container! (Python/Flask) - Visited {count} times.'
    return 'Hello from inside a Docker container! (Python/Flask)'

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

2. requirements.txt



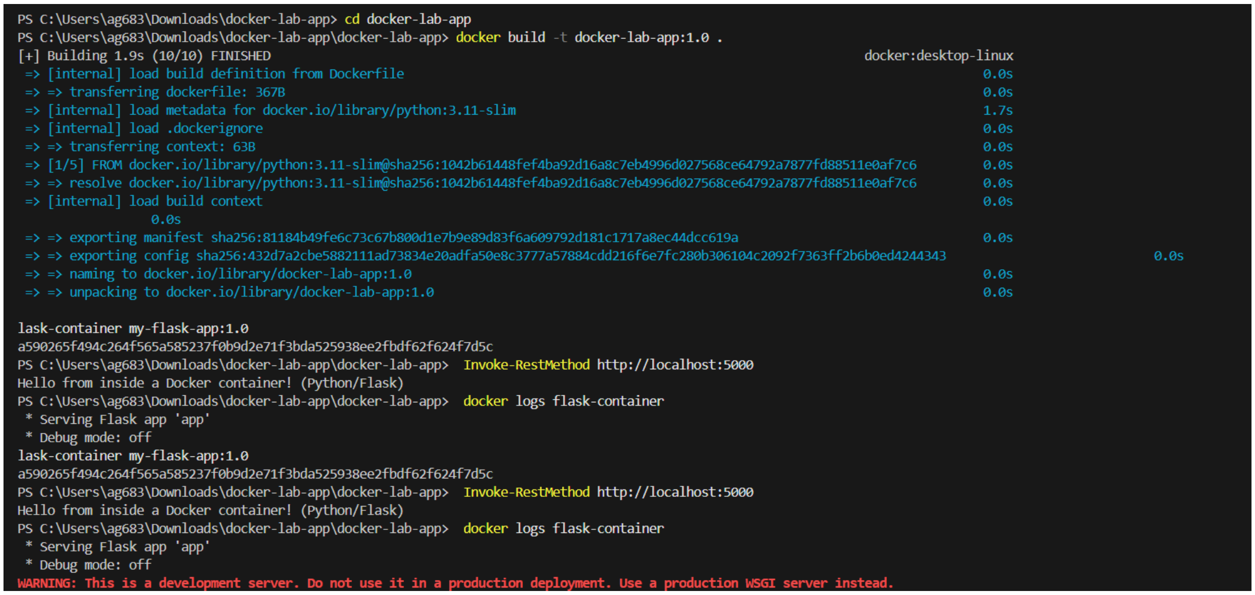
```
flask==3.0.3
redis==5.0.3
```

3. Docker file

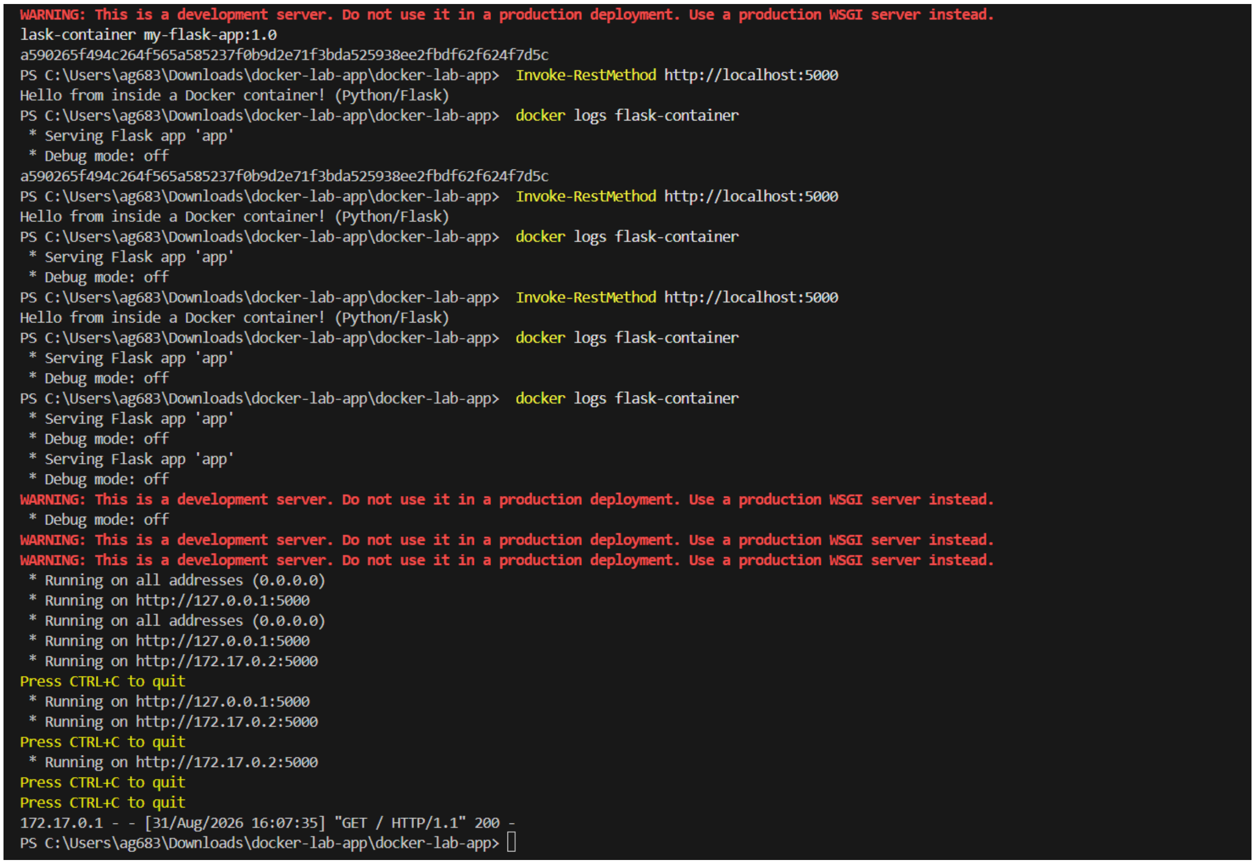


```
# Base image
FROM python:3.11-slim

# Set working directory
WORKDIR /usr/src/app

# Copy dependency manifest first (layer caching)
COPY requirements.txt ./

# Install dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Copy app source
COPY . .

# Expose port
EXPOSE 5000

# Start the app
CMD ["python", "app.py"]
```

4. Build Process & Execution Verification



```
PS C:\Users\ag683\Downloads\docker-lab-app> cd docker-lab-app
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app> docker build -t docker-lab-app:1.0 .
[+] Building 1.9s (10/10) FINISHED                                                       docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                    0.0s
 => => transferring dockerfile: 367B                                                                    0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                     1.7s
 => [internal] load .dockerignore                                                                       0.0s
 => => transferring context: 63B                                                                        0.0s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:1042b61448fef4ba92d16a8c7eb4996d027568ce64792a7877fd88511e0af7c6    0.0s
 => => resolve docker.io/library/python:3.11-slim@sha256:1042b61448fef4ba92d16a8c7eb4996d027568ce64792a7877fd88511e0af7c6    0.0s
 => [internal] load build context                                                                       0.0s
              0.0s
 => => exporting manifest sha256:81184b49fe6c73c67b800d1e7b9e89d83f6a609792d181c1717a8ec44dcc619a       0.0s
 => => exporting config sha256:432d7a2cbe5882111ad73834e20adfa50e8c3777a57884cdd216f6e7fc280b306104c2092f7363ff2b6b0ed4244343                0.0s
 => => naming to docker.io/library/docker-lab-app:1.0                                                   0.0s
 => => unpacking to docker.io/library/docker-lab-app:1.0                                                0.0s

lask-container my-flask-app:1.0
a590265f494c264f565a585237f0b9d2e71f3bda525938ee2fbdf62f624f7d5c
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  Invoke-RestMethod http://localhost:5000
Hello from inside a Docker container! (Python/Flask)
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker logs flask-container
 * Serving Flask app 'app'
 * Debug mode: off
lask-container my-flask-app:1.0
a590265f494c264f565a585237f0b9d2e71f3bda525938ee2fbdf62f624f7d5c
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  Invoke-RestMethod http://localhost:5000
Hello from inside a Docker container! (Python/Flask)
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker logs flask-container
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
```



```
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
lask-container my-flask-app:1.0
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PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker logs flask-container
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WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
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 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
Press CTRL+C to quit
172.17.0.1 - - [31/Aug/2026 16:07:35] "GET / HTTP/1.1" 200 -
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>
```